MATH 201 Your name: _____

Learning target 2.7, version 1

Compute derivatives of implicitly-defined functions.

Consider the curve defined implicitly by the equation

$$y^3 - x \cdot y = -6.$$

1. Use implicit differentiation to find an expression for $\frac{dy}{dx}$ that depends on x and y. Show all of your work and thinking clearly, using correct notation.

2. The point $(7,2)$ lies on this curve. (I promise this is true, but how would you check?)
   Find the exact slope of the tangent line to this curve at the point $(7,2)$. Show clearly how you determined this value, and label your work with proper notation.

3. (Also LT 1.8!) Figure out the equation of the tangent line $L(x)$ to this curve at the point $(7,2)$, and use your tangent line equation to make a reasonable guess about the y-coordinate of the point $(6.8, ____)$.

MATH 201 Your name: _____

Learning target 2.7, version 2

Compute derivatives of implicitly-defined functions.

The equation $x^3 - y^3 = 9xy$ makes a curve called the *folium of Descartes*.

1. Use implicit differentiation to find an expression for $\frac{dy}{dx}$ that depends on x and y. Show all of your work and thinking clearly, using correct notation.

2. Find the exact slope of the tangent line to this curve at the point $(-2,4)$. Show clearly how you determined this value, and label your work with proper notation.

3. (Also AD2) Find an equation for the tangent line to this curve at this point, and use your tangent line to make a reasonable guess about the y-coordinate of the point $(-2.5, ____)$.

MATH 201 Your name: _____

Learning target 2.7, version 3

Compute derivatives of implicitly-defined functions.

Consider the curve defined implicitly by the equation

$$x^5 - 8y^4 = \sin(y) - 7.$$

(a) Use implicit differentiation to find an expression for $\frac{dy}{dx}$ that depends on x and y. Show all of your work and thinking clearly, using correct notation.

(b) This curve contains the point $(-1.4758, 0)$ (or, pretty close to that). Find the exact slope of the tangent line to this curve at this point. Show clearly how you determined this value, and label your work with proper notation.

MATH 201 Your name: _____

Learning target 2.7, version 4

Compute derivatives of implicitly-defined functions.

Consider the curve defined implicitly by the equation

$$\sin(x+y)+\cos(x-y)=1.$$

(a) Use implicit differentiation to find an expression for $\frac{dy}{dx}$ that depends on x and y. Show all of your work and thinking clearly, using correct notation.

(b) This curve contains the point $\left(\frac{\pi}{2},\frac{\pi}{2}\right)$. Find the exact slope of the tangent line to this curve at this point. Show clearly how you determined this value, and label your work with proper notation.